

Batman Attempts

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1 Error Correction

I made two errors in last week's report that I need to correct. First, I confused myself with $f(x)$ and $f(z)$. Here's the correct derivation: Since

$$S_0(x, \theta) = \frac{\nabla_{\theta} g(x, \theta)}{g(x, \theta)}$$

,

$$\begin{aligned} S_{mix}(x, \theta) &= \frac{\nabla_{\theta} h(x, \theta)}{h(x, \theta)} = \frac{(1-p)\nabla_{\theta} g(x, \theta)}{pf(x) + (1-p)g(x, \theta)} \\ &= \frac{(1-p)g(x, \theta)S_0(x, \theta)}{pf(x) + (1-p)g(x, \theta)} \\ &= \beta S_0 \end{aligned}$$

for

$$\beta = \frac{(1-p)g(x, \theta)}{pf(x) + (1-p)g(x, \theta)} = \frac{1-p}{p \frac{f(x)}{f(z)} |\phi_z(x, \theta)| + (1-p)}.$$

I'd like to correct an error that I made last time in the report. Here are the results for $p = 0.1$ and $p = 0.5$. I didn't use the updated variable in my function, which made the results to be the same. (The loss graphs are still wrong I know!)

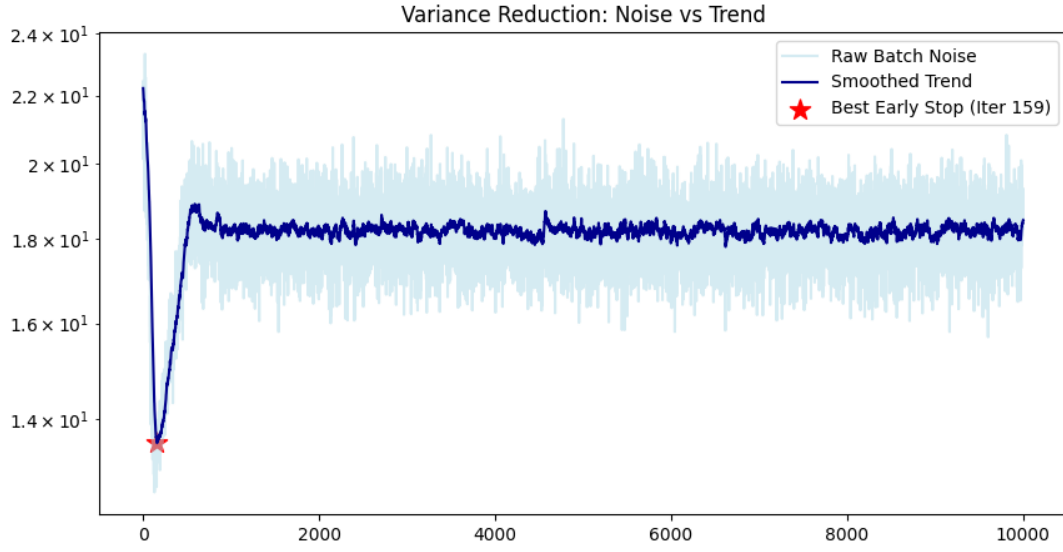


Figure 1: The variance of the neural net after early stopping and mixing ($p = 0.1$).

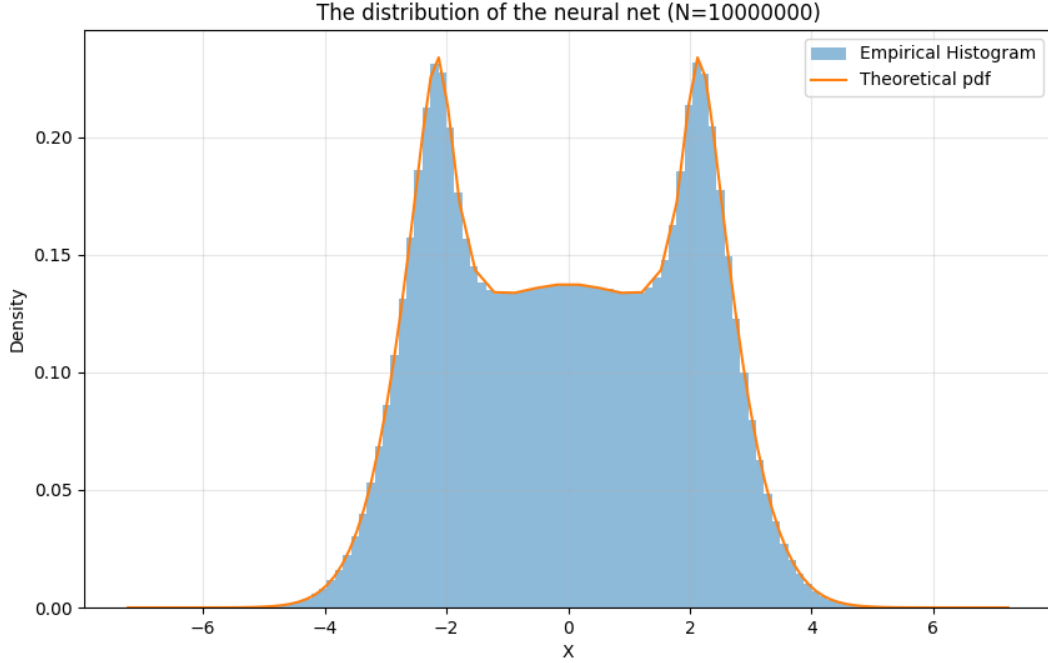


Figure 2: The distribution of the neural net after early stopping and mixing ($p = 0.1$) at $\theta = [1.45, 0.19, 0.65, 0.1]$.

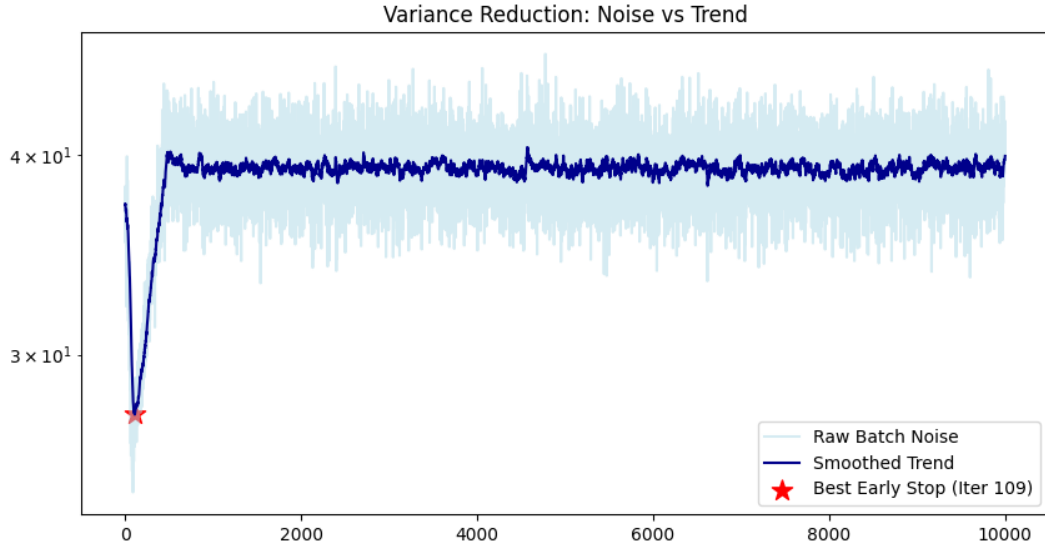


Figure 3: The variance of the neural net after early stopping and mixing ($p = 0.5$).

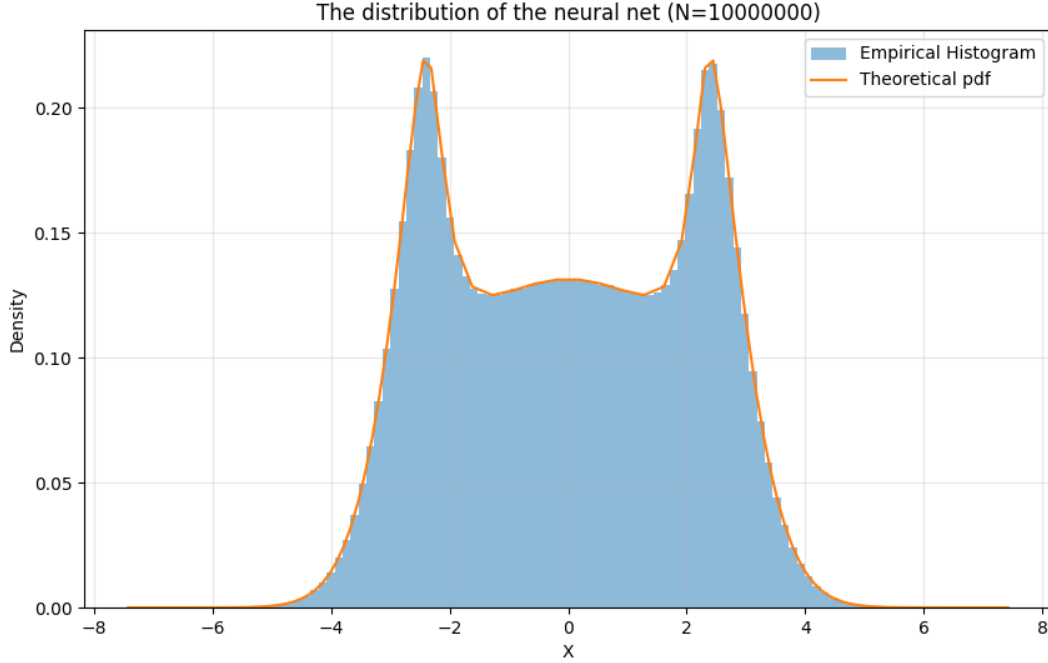


Figure 4: The distribution of the neural net after early stopping and mixing ($p = 0.5$) at $\theta = [1.519, 0.204, 0.699, 0.1]$.

2 Code checking

I'm sorry to inform you this, but after checking I think my code matches the notes. I think

$$g(x, \theta) = \frac{f(z)}{|\phi(z, \theta)|}$$

and the absolute value is necessary instead of statement in the notes.

When ϕ is increasing,

$$P(X \leq x) = P(\phi(Z) \leq x) = P(Z \leq \phi^{-1}(x)),$$

so when we take the derivative we get that

$$g(x) = f(z) \frac{1}{\phi'(\phi^{-1}(x))} = \frac{f(z)}{|\phi_z(z)|}$$

When ϕ is decreasing,

$$P(X \leq x) = P(\phi(Z) \leq x) = P(Z \geq \phi^{-1}(x)) = 1 - P(Z < \phi^{-1}(x)),$$

so when we take the derivative we get that

$$g(x) = -\frac{f(z)}{\phi_z(z)} = \frac{f(z)}{|\phi_z(z)|}.$$

Of course this is assuming that ϕ is monotonically increasing/decreasing.

I did try a few things in an attempt to figure this out. I changed my clip threshold to 10 to prevent very drastic drop in parameters. I also noticed that the increase in loss isn't as drastic when I reduced the trimming rate to 0.05.

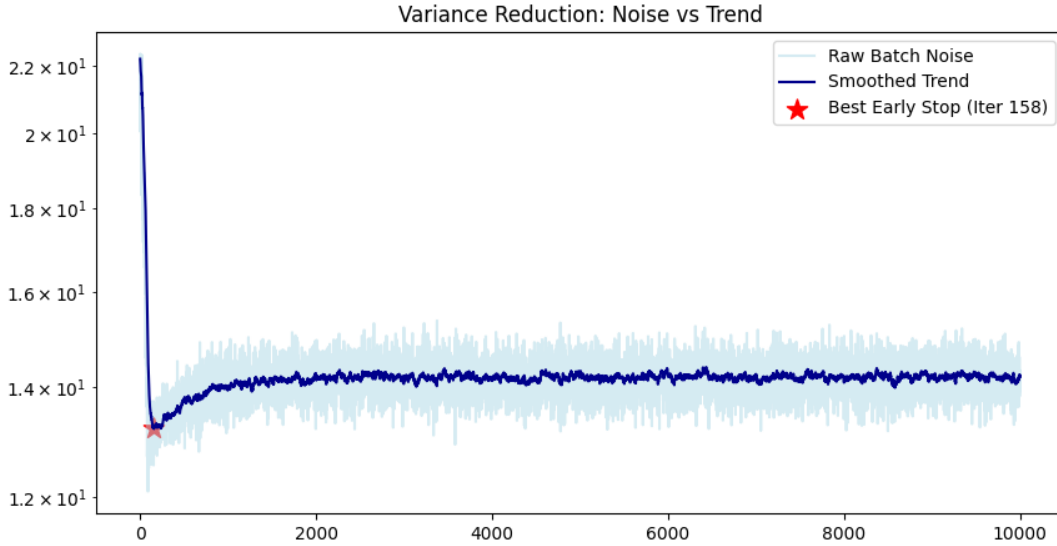


Figure 5: Trimming only the top and bottom 5%.

I also displayed the first 300 iterations where the turning point happens. You can see that the mean gradient loss are completely different before and after trimming and clipping (trimming top and bottom 20%), so I think the estimator might be very biased.

```

Iter   0: Loss=19.4104 | Est=2.6976
        Theta: [1.9983747  0.5006283  0.99886197 0.20025504]
        Raw Grad : [ 8.706446 -2.8515768  5.872943 -3.4941852] (Mag: 11.43)
        Avg Grad : [ 3.2505045 -1.2566428  2.2760668 -0.51005536]
Iter  30: Loss=18.8291 | Est=2.7394
        Theta: [1.9191236  0.53104496 0.94349945 0.21213695]
        Raw Grad : [ 9.8638935 -4.2838287  8.103858 -0.8795383] (Mag: 13.49)
        Avg Grad : [ 4.3502293 -1.9153229  3.345108 -0.14209001]
Iter  60: Loss=18.7244 | Est=2.9809
        Theta: [1.8098602  0.4432349  0.8410972  0.16919167]
        Raw Grad : [ 14.439487 -10.338545  16.396217  11.03643 ] (Mag: 26.57)
        Avg Grad : [ 9.106535 -6.59366  8.659863  3.4126441]
Iter  90: Loss=14.8746 | Est=2.9448
        Theta: [1.6677574  0.23355149 0.71281755 0.1          ]
        Raw Grad : [ 15.181604 -24.647223  21.475296  45.11222 ] (Mag: 57.74)
        Avg Grad : [ 7.7973003 -16.39086  5.8658624  17.007727 ]
Iter 120: Loss=14.6422 | Est=3.2554
        Theta: [1.5551814  0.21103628 0.6460076  0.1          ]
        Raw Grad : [ 4.274379 -12.5211  9.405015  24.55989 ] (Mag: 29.44)
        Avg Grad : [ 5.582189 -21.123127  1.2702001  30.48453 ]
Iter 150: Loss=13.5395 | Est=3.0356
        Theta: [1.481627  0.1963254  0.64715844 0.1          ]
        Raw Grad : [ 9.321261 -31.378752  11.808552  64.983986] (Mag: 73.71)
        Avg Grad : [ 10.499189 -35.88744  7.965821  64.88283 ]
Iter 180: Loss=16.0015 | Est=3.4589
        Theta: [1.4283732  0.18567465 0.6742417  0.1          ]
        Raw Grad : [ -4.750441  5.250907  0.79997635 -11.725866 ] (Mag: 13.72)
        Avg Grad : [ 6.729925 -33.11975  -0.8192936  50.344246 ]
Iter 210: Loss=14.1936 | Est=3.0850
        Theta: [1.359046  0.1718092 0.6758123 0.1          ]
        Raw Grad : [ -9.853462  23.371696  -0.96239084 -39.069855 ] (Mag: 46.59)
        Avg Grad : [ 6.909597 -28.559568  1.3957953  41.875023 ]
Iter 240: Loss=14.0181 | Est=2.9125
        Theta: [1.3125647  0.16251294 0.7026178  0.1          ]
        Raw Grad : [-20.141687  34.815285 -11.353717 -54.436756] (Mag: 68.63)
        Avg Grad : [ -0.49980947 -14.264088  -5.644549  17.604115 ]
Iter 270: Loss=15.2982 | Est=3.2252
        Theta: [1.2762445  0.15524891 0.7295783  0.1          ]
        Raw Grad : [-24.380203  57.585426  -9.423501 -89.69263 ] (Mag: 109.75)
        Avg Grad : [ 2.2547152 -12.4469595  -1.6932825  10.25861 ]
Iter 1000: Loss=15.4891 | Est=2.6446
        Theta: [1.          0.1          0.92852956 0.1          ]

```

Figure 6: Log for the first 300 iterations with the raw and trimmed gradient loss.